

Date	Day	Workstream	Task	Expected Output	Priority	Status	Notes
05/06	Fri	Two-Tower	Commit stabilization/review gate	Clean repo checkpoint	High	Done	No more model features for now
05/06	Fri	Two-Tower	Read experiment summary, code walkthrough, mentor Q&A	Personal understanding notes	High	Done	Prepare to explain project
05/06	Fri	Planning	Finalize weekly Excel plan	Weekly plan ready	High	Done	Date-based plan
05/06	Fri	PySpark	Choose dataset and define problem	Dataset + target column selected	High	Done	Churn/propensity/classification
05/06	Fri	API	Define API mini scope	API endpoint/schema/test plan	High	Done	Prediction lookup by user_id
05/06	Fri	API	Draft API documentation	API_DOCS.md draft	High	Done	Mentor emphasized API docs; draft before implementation, finalize after code
06/06	Sat	PySpark	Do Pandas EDA	5 charts + EDA notes	High	To Do	Quick data understanding
06/06	Sat	PySpark	Define cleaning and feature rules	Feature list + preprocessing plan	High	To Do	Prepare PySpark conversion
06/06	Sat	API	Generate 500k mock prediction rows	Mock prediction dataset	High	To Do	user_id, prediction_score
06/06	Sat	Docs	Write daily progress note	Daily summary	Medium	To Do	Track learning/output
08/06	Mon	PySpark	Convert preprocessing workflow to PySpark	PySpark preprocessing script	High	To Do	Pandas → PySpark
08/06	Mon	API	Build FastAPI lookup endpoint	/prediction/{user_id} works	High	To Do	Return prediction score
08/06	Mon	API	Add not-found handling and metadata	404 response, trace_id, timestamp	High	To Do	Production-style response
08/06	Mon	PySpark	Train Logistic Regression model	Model metrics report	High	To Do	Accuracy/F1/AUC
09/06	Tue	API	Add in-memory cache	Cached lookup working	High	To Do	Compare cached vs non-cached
09/06	Tue	API	Finalize API docs after implementation	Updated API_DOCS.md	Medium	To Do	Endpoint, request, response
09/06	Tue	PySpark	Add second model if feasible	Model comparison table	Medium	To Do	RandomForest/XGBoost optional
09/06	Tue	API	Run load test	Load test report	High	To Do	Latency, throughput, error rate
10/06	Wed	API	Record performance metrics	P50/P95/P99, RPS, error rate	High	To Do	Performance test evidence
10/06	Wed	API	Run stress test	Stress test report	High	To Do	Find breaking point
10/06	Wed	API	Compare cache vs no-cache	Cache impact table	High	To Do	Show latency improvement
10/06	Wed	Privacy	Write PII/privacy notes	PII_PRIVACY_NOTES.md	Medium	To Do	Basic privacy awareness
11/06	Thu	PySpark	Finalize PySpark report	PYSPARK_REPORT.md	High	To Do	EDA + pipeline + model
11/06	Thu	API	Finalize API performance report	API_PERFORMANCE_REPO_RT.md	High	To Do	Load/stress/cache results
11/06	Thu	All	Reproducibility check	Commands verified	High	To Do	Run from clean instructions
12/06	Fri	All	Prepare mentor review package	Summary + links + questions	High	To Do	Ready for async review

Date	Day	Workstream	Task	Expected Output	Priority	Status	Notes
12/06	Fri	All	Send weekly update	Mentor update message	High	To Do	Include blockers/questions
12/06	Fri	Buffer	Fix urgent issues	Bugfix/docs cleanup	Medium	To Do	Based on review/self-check

Milestone ID	Date	Workstream	Milestone	Deliverable	Success Criteria	Status
M1	05/06	Two-Tower	Review package checkpoint	Summary, walkthrough, Q&A	Can explain project to mentor	Done
M2	05/06	PySpark	Dataset selected	Dataset + problem statement	Target column and task are clear	Done
M3	05/06	API	API scope finalized	API mini PRD	Endpoint/schema/cache/test plan defined	Done
M3A	05/06	API	API docs draft completed	API_DOCS.md draft	Endpoint/request/response/errors/cache/test plan documented	Done
M4	06/06	PySpark	EDA completed	5 charts + notes	Understand data and features	To Do
M5	06/06	API	Mock data generated	500k prediction rows	Data ready for API lookup	To Do
M6	08/06	PySpark	PySpark preprocessing completed	PySpark script	Workflow runs in PySpark	To Do
M7	08/06	API	API MVP completed	FastAPI endpoint	Lookup by user_id works	To Do
M8	08/06	PySpark	Basic model trained	Logistic Regression metrics	Model trains/evaluates end-to-end	To Do
M9	09/06	API	Cache added	Cached API	Repeated lookup faster or documented	To Do
M10	09/06	API	Load test completed	Load test report	Latency/RPS/error rate recorded	To Do
M11	10/06	API	Stress test completed	Stress test report	System limit identified	To Do
M12	10/06	Privacy	PII notes completed	Privacy note	Basic PII risks documented	To Do
M13	11/06	PySpark/API	Reports finalized	PySpark + API reports	Mentor-reviewable docs ready	To Do
M14	12/06	All	Weekly package sent	Update message + links	Mentor can review asynchronously	To Do

Date	Day	Must-Have Output	Nice-to-Have	Risk if Missed	Status
05/06	Fri	Two-Tower review checkpoint, PySpark dataset chosen, API scope + API docs draft defined	Clean commits	Next work becomes unclear	Done
06/06	Sat	PySpark EDA + mock prediction data	First API skeleton	PySpark/API delayed	To Do
08/06	Mon	PySpark preprocessing + FastAPI endpoint + Logistic Regression	Basic API tests	No working PySpark/API MVP	To Do
09/06	Tue	API cache + API docs + load test	Second PySpark model	Missing performance evidence	To Do
10/06	Wed	Stress test + cache comparison + PII notes	Charts/tables for report	Missing production-awareness evidence	To Do
11/06	Thu	Final PySpark/API reports	Full rerun commands	Review package incomplete	To Do
12/06	Fri	Mentor update sent	Fix feedback immediately	No review cycle	To Do

Date	Workstream	Risk / Blocker	Impact	Mitigation	Status
05/06	Two-Tower	Low personal understanding due to vibe coding	Hard to explain to mentor	Read walkthrough + Q&A before adding features	Open
05/06	Two-Tower	Popularity baseline still beats two-tower	Model result may look weak	Explain sparsity, cold users, baseline strength	Open
05/06	PySpark	Dataset choice takes too long	Delays MVP	Pick simple classification dataset quickly	Open
06/06	PySpark	PySpark setup issue	Blocks pipeline	Use local PySpark or fallback notebook environment	Open
08/06	API	API scope creep	Delays testing	Keep lookup API simple	Open
09/06	API	Cache overcomplication	Wastes time	Start with in-memory cache only	Open
09/06	API	Load test tool setup slow	Delays performance report	Use simplest available tool first	Open
10/06	Privacy	PII topic too broad	Too much theory	Keep to basic PII/access/logging notes	Open

Status
To Do
In Progress
Blocked
Done
Deferred

Priority
High
Medium
Low